Worksheet 3.1 — Reciprocal & Rational Functions

Grade 12 Advanced Functions (MHF4U) · Unit 3: Rational Functions

The Big Ideas

A rational function $f(x) = \dfrac{p(x)}{q(x)}$ has vertical asymptotes at zeros of q that do not cancel, a horizontal asymptote from the degrees, and a hole wherever a factor cancels. In every step, write the **whole function**, even when you are only working on its numerator or denominator.

- Vertical asymptote: set the denominator to 0 and check the numerator is not 0 there. The domain excludes every such x.
- Horizontal asymptote: lower top $\Rightarrow y = 0$; equal degrees $\Rightarrow$ ratio of leading coefficients; higher top $\Rightarrow$ none.
- Hole: factor, cancel, then substitute the cancelled x into the simplified function to get its height.

Core Concepts

Reciprocal functions

The reciprocal $y = \dfrac{1}{f(x)}$ blows up where $f(x) = 0$, creating a **vertical asymptote**, and flattens toward 0 where f grows large.

Rational functions

A **rational function** is a ratio of polynomials $y = \dfrac{p(x)}{q(x)}$; its behaviour is governed by the zeros of q (the denominator) and by the degrees of p and q. Always **factor first**: a factor that cancels gives a hole, a factor that stays in the denominator gives a vertical asymptote.

Asymptotes

Vertical asymptotes sit where the denominator is zero but the numerator is not. For the horizontal asymptote compare degrees: top lower $\Rightarrow y = 0$; equal degrees $\Rightarrow$ ratio of leading coefficients; top higher $\Rightarrow$ none (a slant asymptote if exactly one degree higher).

Writing solutions

In every step write the **whole function**, even when you are only working on the numerator or the denominator. For example: "for $y = \dfrac{x+1}{x-3}$, set the denominator $x - 3 = 0$, so $x = 3$." This keeps clear which function you are analysing.

Worked Examples

Example 1 — Vertical asymptote

Vertical asymptote of $y = \dfrac{1}{x-3}$?

Solution. **Step 1:** Set the denominator of $y = \dfrac{1}{x-3}$ equal to zero: $x - 3 = 0 \Rightarrow x = 3$.

Step 2: At $x = 3$ the numerator of $y = \dfrac{1}{x-3}$ is $1 \neq 0$ and nothing cancels, so the function is undefined there and the curve shoots off to $\pm\infty$.

Answer: for $y = \dfrac{1}{x-3}$ the vertical asymptote is $x = 3$ (domain $x \neq 3$).

Example 2 — Horizontal asymptote

Horizontal asymptote of $y = \dfrac{1}{x-3}$?

Solution. **Step 1:** In $y = \dfrac{1}{x-3}$ the numerator 1 has degree 0 and the denominator $x-3$ has degree 1.

Step 2: Since $0 < 1$, the horizontal asymptote of $y = \dfrac{1}{x-3}$ is $y = 0$. Check: at $x = 1000$, $y = \dfrac{1}{997} \approx 0.001$, very close to 0.

Answer: $y = 0$.

Example 3 — The reciprocal graph

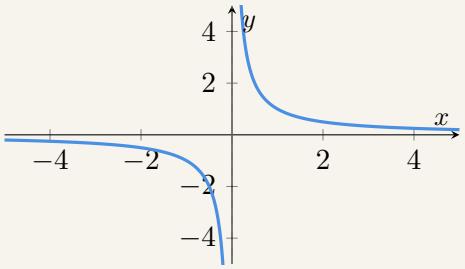
Describe $y = \dfrac{1}{x}$.

Solution. **Step 1 (vertical asymptote):** Set the denominator of $y = \dfrac{1}{x}$ equal to zero: $x = 0$. The numerator $1 \neq 0$, so $x = 0$ is a VA.

Step 2 (horizontal asymptote): In $y = \dfrac{1}{x}$ the numerator has degree 0 and the denominator x has degree 1, so the HA is $y = 0$.

Step 3 (signs): For $y = \dfrac{1}{x}$: if $x > 0$ then $y > 0$ (quadrant I); if $x < 0$ then $y < 0$ (quadrant III). The branches lie in opposite quadrants.

Step 4 (intercepts): $y = \dfrac{1}{x}$ can never equal 0 (the numerator is 1), and $x = 0$ is not allowed, so there are no intercepts:



Example 4 — Domain

Domain of $y = \dfrac{1}{x+2}$?

Solution. **Step 1:** The domain of $y = \dfrac{1}{x+2}$ excludes every x that makes the denominator zero: $x + 2 = 0 \Rightarrow x = -2$.

Answer: all real numbers except -2, i.e. $x \neq -2$ (and $x = -2$ is a vertical asymptote of $y = \dfrac{1}{x+2}$).

Example 5 — Equal degrees

Horizontal asymptote of $y = \dfrac{2x}{x-1}$?

Solution. **Step 1:** In $y = \dfrac{2x}{x-1}$ the numerator $2x$ has degree 1 and the denominator $x - 1$ has degree 1 — equal degrees.

Step 2: The HA of $y = \dfrac{2x}{x-1}$ is the ratio of the leading coefficients: $y = \dfrac{2}{1} = 2$. Check: at $x = 1000$, $y = \dfrac{2000}{999} \approx 2.002$.

Answer: $y = 2$.

Example 6 — A hole

Describe $y = \dfrac{x^2 - 25}{x - 5}$.

Solution. **Step 1 (factor):** The numerator of $y = \dfrac{x^2 - 25}{x - 5}$ is a difference of squares: $x^2 - 25 = (x - 5)(x + 5)$, so $y = \dfrac{(x-5)(x+5)}{x-5}$.

Step 2 (cancel): The factor $x - 5$ cancels in $y = \dfrac{(x-5)(x+5)}{x-5}$, leaving $y = x + 5$ for $x \neq 5$.

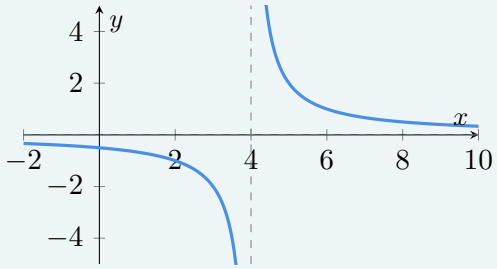
Step 3 (hole): Substitute $x = 5$ into the simplified $y = x + 5$: $y = 10$. The hole is at $(5, 10)$, and there is no vertical asymptote at $x = 5$ because the factor cancelled.

Answer: $y = \dfrac{x^2 - 25}{x - 5}$ is the line $y = x + 5$ with a hole at $(5, 10)$.

Example 7 — Shifted asymptote

Vertical asymptote of $y = \dfrac{2}{x-4}$, graphed:

Solution. **Step 1:** Set the denominator of $y = \dfrac{2}{x-4}$ equal to zero: $x-4=0 \Rightarrow x=4$. The numerator is $2 \neq 0$, so the VA is $x=4$ (dashed).
Step 2: In $y = \dfrac{2}{x-4}$ the numerator has degree $0 < 1$, so the HA is $y=0$ (dashed):



Example 8 — Equal degrees

Horizontal asymptote of $y = \dfrac{7x}{x-2}$?

Solution. **Step 1:** In $y = \dfrac{7x}{x-2}$ both the numerator $7x$ and the denominator $x-2$ have degree 1.
Step 2: The HA of $y = \dfrac{7x}{x-2}$ is $y = \dfrac{7}{1} = 7$.
Answer: $y=7$ (the vertical asymptote of $y = \dfrac{7x}{x-2}$ is $x=2$).

Example 9 — Domain

Domain of $y = \dfrac{x}{x+7}$?

Solution. **Step 1:** The domain of $y = \dfrac{x}{x+7}$ excludes the x that makes the denominator zero: $x+7=0 \Rightarrow x=-7$.
Step 2: At $x=-7$ the numerator of $y = \dfrac{x}{x+7}$ is $-7 \neq 0$, so it is also a vertical asymptote.
Answer: $x \neq -7$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Vertical asymptote of $y = \dfrac{1}{x-8}$?

Question 2

Horizontal asymptote of $y = \dfrac{6}{x-2}$?

Question 3

Domain of $y = \dfrac{x+3}{x-9}$?

Question 4

Horizontal asymptote of $y = \dfrac{9x}{x-1}$?

Question 5

Where is the hole in $y = \dfrac{x^2-49}{x-7}$?

Question 6

Vertical asymptote of $y = \dfrac{2}{x+3}$?

Question 7

Horizontal asymptote of $y = \dfrac{4x}{x-7}$?

Question 8

Domain of $y = \dfrac{1}{x-1}$?

Question 9

Horizontal asymptote of $y = \dfrac{1}{x^2+1}$?

Question 10

Vertical asymptote of $y = \dfrac{x+1}{x-6}$?

Question 11

Simplify $y = \dfrac{x^2-9}{x+3}$ and state any hole.

Question 12

Horizontal asymptote of $y = \dfrac{3x}{2x-1}$?

Question 13 — Challenge

State the VA, HA and domain of $y = \dfrac{2x}{x-4}$.

Question Recap & Answer Key

Question Recap

1. Vertical asymptote of $y = \dfrac{1}{x-8}$?
2. Horizontal asymptote of $y = \dfrac{6}{x-2}$?
3. Domain of $y = \dfrac{x+3}{x-9}$?
4. Horizontal asymptote of $y = \dfrac{9x}{x-1}$?
5. Where is the hole in $y = \dfrac{x^2-49}{x-7}$?
6. Vertical asymptote of $y = \dfrac{2}{x+3}$?
7. Horizontal asymptote of $y = \dfrac{4x}{x-7}$?
8. Domain of $y = \dfrac{1}{x-1}$?
9. Horizontal asymptote of $y = \dfrac{1}{x^2+1}$?
10. Vertical asymptote of $y = \dfrac{x+1}{x-6}$?
11. Simplify $y = \dfrac{x^2-9}{x+3}$ and state any hole.
12. Horizontal asymptote of $y = \dfrac{3x}{2x-1}$?
13. State the VA, HA and domain of $y = \dfrac{2x}{x-4}$.

✓ Answer Key

1. For $y = \dfrac{1}{x-8}$: $x-8=0 \Rightarrow x=8$ (numerator $1 \neq 0$), so the VA is $x=8$
2. For $y = \dfrac{6}{x-2}$: numerator degree $0 <$ denominator degree 1, so the HA is $y=0$
3. For $y = \dfrac{x+3}{x-9}$: $x-9=0 \Rightarrow x=9$ is excluded, so the domain is $x \neq 9$
4. For $y = \dfrac{9x}{x-1}$: equal degrees, $\dfrac{9}{1}=9$, so the HA is $y=9$
5. $y = \dfrac{x^2-49}{x-7} = \dfrac{(x-7)(x+7)}{x-7} = x+7$ for $x \neq 7$; the hole is at $x=7$, i.e. $(7,14)$
6. For $y = \dfrac{2}{x+3}$: $x+3=0 \Rightarrow x=-3$ (numerator $2 \neq 0$), so the VA is $x=-3$
7. For $y = \dfrac{4x}{x-7}$: equal degrees, $\dfrac{4}{1}=4$, so the HA is $y=4$ (and the VA is $x=7$)
8. For $y = \dfrac{1}{x-1}$: $x-1=0 \Rightarrow x=1$ is excluded, so the domain is $x \neq 1$
9. For $y = \dfrac{1}{x^2+1}$: numerator degree $0 <$ denominator degree 2, so the HA is $y=0$ (and $x^2+1 \neq 0$, so there is no VA)
10. For $y = \dfrac{x+1}{x-6}$: $x-6=0 \Rightarrow x=6$ (numerator $6+1=7 \neq 0$), so the VA is $x=6$
11. $y = \dfrac{x^2-9}{x+3} = \dfrac{(x-3)(x+3)}{x+3} = x-3$ for $x \neq -3$; the hole is at $x=-3$, i.e. $(-3,-6)$
12. For $y = \dfrac{3x}{2x-1}$: equal degrees, $\dfrac{3}{2}$, so the HA is $y=\frac{3}{2}$ (and the VA is $x=\frac{1}{2}$)
13. For $y = \dfrac{2x}{x-4}$: VA $x-4=0 \Rightarrow x=4$ (numerator $8 \neq 0$); HA equal degrees, $\dfrac{2}{1}=2$, so $y=2$; domain $x \neq 4$